

Olympics Data Analysis

Business Problem Statement

The Olympic Games generate a large amount of data related to athletes, sports, countries, events, and medal achievements. Understanding this data can help identify participation trends, popular sports, top-performing countries, and athlete characteristics.

Sports analysts and researchers are interested in exploring historical Olympic records to understand how participation has changed over time, which countries have performed best, and what factors are associated with medal-winning athletes.

You are tasked with analyzing the Olympic athlete dataset to answer the following business question:

“How can Olympic data be used to identify participation trends, analyze country performance, understand athlete demographics, and uncover insights about medal achievements?”

Deliverables

1. Data Preparation & Cleaning (Python)

- Load and clean the Olympic dataset.
- Handle missing values and prepare data for analysis.

2. Data Analysis (SQL)

- Store data in PostgreSQL.
- Perform SQL queries to analyze participation, sports popularity, country performance, and medal distribution.

3. Visualization & Insights (Power BI)

- Build an interactive dashboard to visualize key trends and patterns.

4. Report & Presentation

- Summarize findings and insights in a project report.
- Present important results using charts and dashboard visuals.

5. GitHub Repository

- Include Python scripts, SQL queries, dashboard files, and project documentation in a structured repository.